

1912

1913

“IT IS PERHAPS... INDELICATE TO ASK... MOTHER EARTH HER AGE, BUT SCIENCE ACKNOWLEDGES NO SHAME AND... HAS... ATTEMPTED TO WREST FROM HER A SECRET WHICH IS PROVERBIALY WELL GUARDED.”

Arthur Holmes, British geologist, from *The Age of the Earth*, 1913

The archaeologist Charles Dawson (left) convinced paleontologist Arthur Smith-Woodward (center) that he had found the ape-human missing link.

ON FEBRUARY 14, British archaeologist **Charles Dawson** (1864–1916) uncovered **primate skull fragments and a jaw** at the **Pittdown** gravel pit, England. They were reported to be the missing link between humans and apes and attracted much interest from experts at the British Museum of Natural History. More than 40 years later, in 1953, **Pittdown Man** was shown to be a hoax—it was in fact fragments of a modern human and an orang-utan.

This year also saw the dawn of a new way of analyzing chemical structures. German physicists **Walter Friedrich** (1874–1958) and **Paul Knipping** (1883–1935) showed that when X-rays were scattered through a crystal, the resulting diffraction pattern recorded on a photographic plate could be used to **identify the positions of the crystal's atoms**. Later, German physicist **Max von Laue** (1879–1960) explained the theory for this scattering behavior and used it to show that X-rays were **not particles, but waves with very short wavelengths**. X-ray diffraction would be developed as a mainstay of analytical chemistry. British biochemist **Frederick Hopkins** (1861–1947) was experimenting with the diets of animals and announced that, along with carbohydrates,

proteins, and fats, a healthy diet needed **accessory food factors**. Polish-born biochemist **Casimir Funk** (1884–1967) proposed a name for them—**vitamines**, from “vital” and “amine.” He thought that they belonged to a class of chemicals called amines, but when it was shown that this was not the case, the name was changed to **vitamin**.

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THE SPEED THE ANDROMEDA GALAXY MOVES AWAY FROM THE SUN

American astronomer **Vesto Slipher** (1875–1969), using a technique of light analysis called spectroscopy (see 1860), observed a shift toward **red wavelengths for light** coming from the **Andromeda galaxy**. This **redshift** indicated that the galaxy was moving away. Slipher's studies provided early evidence for the idea that **the Universe was expanding**.

BRITISH PHYSICIST FREDERICK SODDY had worked with Ernest Rutherford on the nature of radioactivity and identification of decay products. He found that these products had different atomic masses, but held positions in the periodic table that appeared to be occupied by known elements. Soddy concluded that they must be varieties of the elements, and in 1913 he called them **isotopes**.

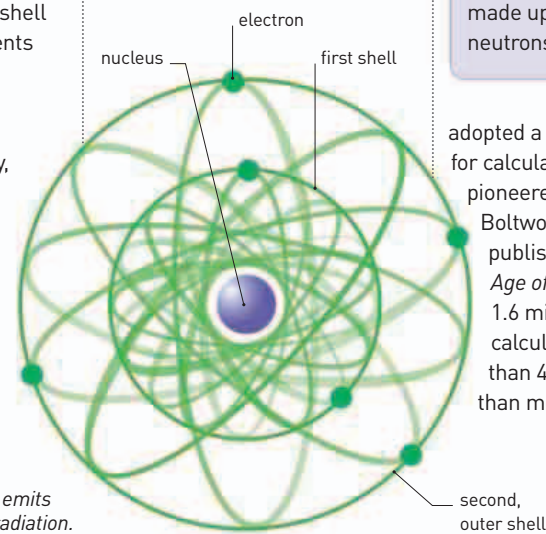
Danish physicist **Niels Bohr** (1885–1962) **modified Rutherford's nuclear model of the atom** to take into account new ideas from quantum physics. Bohr proposed that electrons orbiting the atom's nucleus occupied various shells or energy levels called orbitals. The electrons in the outer shell determined the way elements reacted chemically.

At the BASF (*Badische Anilinund Soda-Fabrik*) chemical plant in Germany, chemist Carl Bosch oversaw the **first industrial production of ammonia** for use in agricultural fertilizers. Bosch had modified a

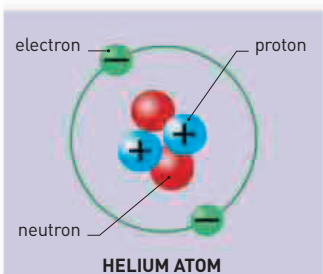
chemical process first described by Fritz Haber (see 1905)—now known as **the Haber-Bosch process**. The factory would reach full production capacity in 1914.

In December, British physicist **Henry Moseley** (1887–1915) found that elements emit characteristic X-rays of wavelengths determined by the numbered position in the periodic table. His discovery supported Van den Broek's idea that the position had physical meaning in the form of the **number of positive charges in an element's atomic nucleus**—its atomic number.

British geologist **Arthur Holmes** (1890–1965) had



Bohr's atomic model
Bohr's model suggested electron “shells” around the nucleus. As electrons move between the shells, the atom emits or absorbs electromagnetic radiation.



ATOMIC NUMBER

Henry Moseley proved that the vaguely defined concept of atomic number related to a measurable physical property of the atomic nucleus, later found to be the number of protons. For a neutral atom, this equals the number of encircling electrons. The nucleus is made up of protons and neutrons (neutral particles).

adopted a radiometric technique for calculating **the age of rocks** pioneered by Bertram Boltwood (see 1907). Holmes published his results in *The Age of the Earth*, but—at 1.6 million years—his calculation was still more than 4,000 times shorter than modern results.

January 6 German geophysicist **Alfred Wegener** describes his theory that modern continents have split from an **ancient supercontinent**

June 8 Max von Laue shows that X-rays are waves, not particles

September 17 Vesto Slipher demonstrates Andromeda Galaxy is moving

July 15 Frederick Hopkins states that a healthy diet needs **accessory food factors**

November 11 Lawrence Bragg explains how crystals diffract X-rays

January American geneticist **Henry Sturtevant** describes first **chromosome map**

February 18 Frederick Soddy coins the term **isotope**

April 5 Niels Bohr describes his **quantum model of atomic structure**

April 12 British Ecological Society founded; first body devoted to study of ecology

June 18 British scientist **John Ernest Williamson** took first underwater ocean photographs

Arthur Holmes publishes *The Age of the Earth*

December 9 Henry Moseley shows how an element's **atomic number** relates to the X-rays it emits